opto-sim Research Roadmap

Physical-Layer Fiber-Optic Simulator

A tiered publication plan with effort estimates, experimental-validation requirements, and scope boundaries.

Perspective

opto-sim is a **general-purpose, open-source, validated physical-layer fiber-optic simulator**. The complex-envelope electric field ($\overline{|\mathcal{E}|^2}$ = Watts) is the single source of truth. BB84 QKD is one downstream application protocol, not the defining purpose.

This roadmap is organised around **four papers**. Tier 1 validates the simulator itself against published experimental data. Tiers 2–4 are QKD-focused and depend on Tier 1 being published.

Publication Plan

#	Paper	Target	When	Depends on
A	Software validation: validated physical-layer simulator	OE / JLT / SoftwareX	Now	—
B	QKD environmental sensitivity: temperature, bend, channel	OE / JLT / EPJ QT	After A	A
C	Physical-layer security: side-channel analysis	IEEE T-IFS / NPJ QI / PRX Q.	After B	A+B
D	Full QKD system: decoy-state, SKR, release	Nature Comms / PRX Q.	Optional	A+B+C

Tier 0 — Infrastructure (COMPLETE)

Effort: 1–2 weeks	Deliverable: pytest suite + seeded reproducibility + 48 tests	Paper: None (scaffolding)

Status: All done. No further work planned.

Tier 1 — Software Validation Paper

Goal: Publish a validated physical-layer fiber-optic simulator that reproduces published experimental data across all components.

Novelty: No open-source simulator validates CD, PMD, birefringence, attenuation, laser RIN, phase noise, MZM transfer, and APD response against *both* analytic theory and published experimental/datasheet curves.

1.1 Fiber Channel (DONE)

Validation	Reference	Error	Script
CD: Gaussian pulse broadening	Agrawal Fig. 2.6	0.0000%	`validate_cd.py`
PMD: DGD histogram vs Maxwellian	Razavi Fig. 2.11	KS $p = 0.31$	`validate_pmd.py`
Attenuation: $\alpha = 0.182$ dB/km @ 1550 nm	SMF-28 spec	0.0000%	`validate_attenuation.py`
Birefringence: $\Delta n \propto (r_f/R)^2$	Yuan Fig. 1	0.0000%	`validate_birefringence.py`

These prove the implementation matches the underlying analytic formula to machine precision. They are *necessary* but not *sufficient* for a software paper—they verify coding correctness, not physical fidelity against real-world data.

1.2 Fiber — Experimental Overlay (NEW)

D(λ) dispersion curve — Corning SMF-28 datasheet

What to add	Sellmeier coefficients for fused silica (Malitson 1965) $\to n(\lambda) \to \mathrm{d}^2 n/\mathrm{d}\lambda^2 \to D_{\text{material}}(\lambda) + D_{\text{waveguide}}$
Effort	~2 sessions
Good enough	$\lvert D_{\text{sim}} - D_{\text{datasheet}} \rvert < 1$ ps/(nm·km) across O–L bands
Too much	Full waveguide mode solver; temperature-dependent Sellmeier
Priority	**High** — $D(\lambda)$ is the standard fiber datasheet curve

$\alpha(\lambda)$ attenuation spectrum — Corning SMF-28 datasheet

What to add	Rayleigh (C_R/λ^4) + IR tail + UV tail + OH^- peak at 1383 nm
Effort	~1 session
Good enough	$\lvert \alpha_{\text{sim}} - \alpha_{\text{datasheet}} \rvert < 0.02$ dB/km outside OH peak
Too much	Temperature-dependent α; radiation-induced loss; bend-loss spectrum
Priority	**High** — attenuation spectrum is the second standard fiber curve

1.3 Laser Validation (NEW)

RIN spectrum — $S_{\mathbf{RIN}}(f)$ vs relaxation-oscillation model

What to add	Validation script: RIN time series $\rightarrow$ Welch PSD $\rightarrow$ Coldren Eq. 5.3.38 overlay
Effort	$\sim$1 session
Good enough	Peak frequency $\pm 10\%$; roll-off $1/f^2$; DC level matches RIN_0
Too much	Shot-noise-calibrated high-resolution RIN; $1/f$ noise
Priority	**High** — relaxation-oscillation model is a unique simulator feature

Phase noise — Lorentzian linewidth verification

What to add	Simulate field $\rightarrow$ complex-envelope PSD $\rightarrow$ fit Lorentzian, compare FWHM to input $\Delta\nu$
Effort	$\sim$1 session
Good enough	Fitted FWHM within $\pm 10\%$ of input linewidth
Too much	Gaussian line-shape decomposition; $1/f$ divergence
Priority	**High** — linewidth validation is essential

1.4 MZM Validation (NEW)

Validation	What to add	Effort	Good enough
V_π from crystal params	Report for default geometry	None (test exists)	$\pm 5\%$ of literature
Extinction ratio	Plot E_{out} vs V	None (test exists)	Matches user ER
Push-pull vs single-drive chirp	Compare output phase	$\sim$1 session	Chirp $\propto$ $0.5V_\pi$

Priority: Medium. Well-tested model; one validation script to publish.

1.5 APD Validation (NEW)

Validation	What to add	Effort	Good enough
Responsivity $R(\lambda)$	Spectral $\eta(\lambda)$ for InGaAs; plot vs datasheet	$\sim$1 session	Peak $\pm 10\%$, shape matches
Shot-noise scaling	$I^2_{\mathrm{noise}} = 2eIB \cdot F$	None (test exists)	Numerical tolerance

Priority: Medium.

1.6 System Demonstration

Demonstration	Status	Effort
BB84 QBER under CD+PMD vs distance	Existing graphic	0
Full link budget: laser $\rightarrow$ fiber $\rightarrow$ detector	New	$\sim$1 session

Priority: Medium — completes the system story.

1.7 Tier 1 — Scope Boundaries

IN SCOPE

- Fiber: $D(\lambda)$ dispersion curve, $\alpha(\lambda)$ attenuation spectrum (Sellmeier + Rayleigh + OH⁻)
- Laser: RIN spectrum plot, Lorentzian linewidth verification
- MZM: V_π, extinction, chirp-mode validation scripts
- APD: Spectral responsivity, shot-noise validation
- System: BB84 QBER demonstration, link budget

OUT OF SCOPE

- Nonlinear effects (Kerr, SPM, XPM, FWM) — 3+ months, separate paper
- EDFA / optical amplifier — not needed for passive-fiber validation
- PDL — second-order effect, no reference data
- Splice / connector loss — negligible publication value
- Temperature-dependent Sellmeier — belongs in Tier 2
- Waveguide mode solvers — defeats the purpose of a macroscopic simulator
- More protocols (E91, MDI-QKD, TF-QKD) — strong diminishing returns
- Quantum security analysis — belongs in Tier 3

Effort Summary

Sub-tier	Sessions	New code	Status
1.1 Fiber (analytic)	—	—	✓
1.2 Fiber (experimental)	$\sim$3	Sellmeier, $\alpha(\lambda)$	○
1.3 Laser	$\sim$2	RIN/linewidth scripts	○
1.4 MZM	$\sim$1	Validation script	○
1.5 APD	$\sim$1	Spectral $\eta(\lambda)$ model	○
1.6 System	$\sim$1	Link budget script	○
Total	$\sim$**8 sessions**		

Estimated calendar time: **3–4 weeks** combined coding + paper writing.

Tier 2 — QKD Environmental Sensitivity (DEFERRED)

Requires: Tier 1 published first.
Goal: Publish a journal paper on how temperature and bend radius affect QKD system performance.

Tasks

1. Temperature sweep (0–50 °C, 25 km spans) $\rightarrow \Delta n(T)$, beat length, QBER
2. Bend sensitivity ($R = 10$–100 mm) $\rightarrow \Delta n(R)$, coupled $T \times R$ regimes
3. Time-varying channel (1 °C/min ramp) $\rightarrow$ QBER time series
4. Aerial vs buried comparison (ASHRAE weather data) $\rightarrow$ QBER variance

Scope Boundaries

IN	OUT
Temperature-dependent birefringence and phase drift	Temperature-dependent Sellmeier
Static and slow-ramp temperature profiles	Thermo-mechanical stress modeling
Single-span fiber (25 km)	Multi-span with amplifiers
BB84 phase-encoding	Other protocols
Bend-induced birefringence	Bend-induced attenuation

Effort: $\sim$**6 weeks.** Stop if QBER delta $< 0.5\%$.

Tier 3 — Physical-Layer Security Analysis (DEFERRED)

Requires: Tiers 1 and 2 published.
Goal: Publish a high-impact paper on a QKD side channel that *only a physical-layer simulator can reveal*.

Tasks

1. Laser-noise side-channel (RIN resonance $\rightarrow$ amplitude leakage)
2. Finite-key effects with correlated noise (non-i.i.d. RIN/phase noise)
3. Detector side-channel (dead time, afterpulsing, time-shift attacks)
4. Phase modulator flaw analysis (drive ripple $\rightarrow$ deterministic phase errors)

Effort: $\sim$**5 months.** One strong result beats a catalogue of re-runs.

Tier 4 — Full QKD System Optimisation (OPTIONAL)

Requires: Tiers 1–3.
Goal: Decoy-state protocol, SKR optimisation, open-source release.
Effort: $\sim$**10 months.** Strongly diminishing returns after Tier 3.

Summary

Tier	What	Effort	Paper target	Dependency
0	Infrastructure	DONE	—	—
1	**Software validation**	**3–4 w**	**OE / JLT / SoftwareX**	—
2	QKD environmental	6 w	OE / JLT / EPJ QT	Tier 1
3	QKD security	5 m	IEEE T-IFS / NPJ QI / PRX Q.	Tiers 1–2
4	Full QKD system	10 m	Nature Comms / PRX Q.	Tiers 1–3

Recommended Route

1. **Tier 1 now** (3–4 weeks). Publish a validated physical-layer simulator paper. This is the immediate priority.
2. **If Tier 2's QBER signal is strong** ($\geq 0.5\%$ delta), publish an environmental journal paper (6 weeks).
3. **Tier 3** (5 months). The laser-noise side-channel is the highest-impact result this platform enables.
4. **Stop** after Tier 3 unless you want a flagship capstone. Diminishing returns past this point are steep.

Competitive Landscape

Advantage	opto-sim	NetSquid	qkdX	Comsis	Caputo
Physical-layer field	✓	—	—	✓	✓
Multi-experiment validated	*in prog.*	—	—	—	—
Open-source	✓	✓	—	—	—
Modern (Python + NumPy)	✓	✓	—	—	✓
Laser RIN + phase noise	✓	—	—	—	—
Temp/bend birefringence	✓	—	—	—	—
FFT-based CD + PMD	✓	—	—	—	—
APD with excess noise	✓	—	—	—	—
Seeded reproducibility	✓	—	—	—	—

This roadmap positions opto-sim as the first **open-source, multi-experiment-validated, physical-layer fiber-optic simulator** — a unique position in the landscape.